Assign1

Group Name

G 33

Instructions

The assignment will be done in **teams of 4 students**. The goal is to apply deep learning methods on a research-oriented problem. The assignment description will be available on 28th April 2026. For technical/programming questions, you can go to the Computer Lab and talk with your TA (Optional).

You will submit maxium **3 pages double column paper** using the Latex or Word template here: https://www.ieee.org/conferences/publishing/templates.html The final report will include the following:

1. Title, Authors, Abstract
2. Introduction: Introduction to the problem
3. Used Models: Briefly explain the models used and why you chose them.
4. Approach: Methodology, technical details and choices made
5. Results: Qualitative and quantitative analysis
6. Conclusion: Summary of your findings
7. References

Submissions

The final report (pdf file) need to be submitted to Brightspace by **May 8th 23:59 hrs**. (A github link to the code should be sent as supplementary material but it will not be graded). Please note that I have activated the multiple attempts option which allows you to submit maximum three times and only the last submission will be graded. The grading will be based on the clarity of the document, the technical content, critical analysis and the performance.

Winners and Bonus:

The top three highest-performing groups will be invited to present their results in class on **May 27th**. Each winning group will receive a 0.25 bonus added to their final grade.

Further details

For neural network implementation, you may use any suitable library. We suggest PyTorch or Tensorflow (note that TF v2 incorporates the keras api). The documentation on the official pages of these frameworks includes several tutorials. Some other useful links, which include pointers to model checkpoints and datasets.

- https://github.com/ChristosChristofidis/awesome-deep-learning
- https://paperswithcode.com/

Due on 08 May 2026 23:59

Available on 28 April 2026 23:59. **Access restricted before availability starts.**

Attachments

Xtrain.mat (1,13 KB)

DL_assignment.pdf (96,52 KB)